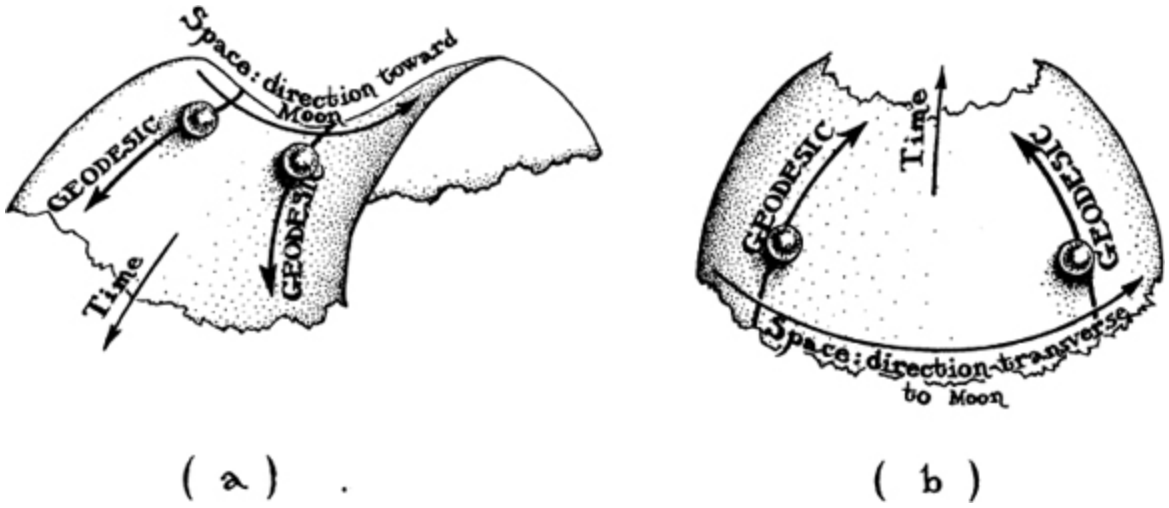


2.6 Two two-dimensional pieces of curved spacetime, in the vicinity of the Earth. The curvature is produced by the Moon. The curvature creates a tidal stretch along the direction toward the Moon (a), and a tidal squeeze along the direction transverse to the Moon (b), and this stretch and squeeze produce the ocean's tides in the manner discussed in Box 2.5, above.

Einstein was a professor in Prague in the summer of 1912, when he realized that tidal gravity and spacetime curvature are one and the same thing. It was a wonderful revelation—though he was not yet certain of it and did not yet understand it as fully as I have described it, and it did not provide a complete explanation of gravity. It told Einstein that spacetime curvature dictates the motion of free particles and raises the tides on the ocean, but it did not tell him how the curvature is produced. Einstein believed that the matter inside the Sun and Earth and other planets is somehow responsible for the curvature. But how? *How does matter warp spacetime, and what are the details of the warpage?* A quest for the *law of warpage* became Einstein's central concern.

A few weeks after "discovering" spacetime curvature, Einstein moved from Prague back to Zurich, to take up a professorship at his alma mater, the ETH. Upon arriving in Zurich in August 1912, Einstein sought advice from an old classmate, Marcel Grossmann, who was now a professor of mathematics there. Einstein explained his idea that tidal gravity is spacetime curvature, and then asked whether any mathematician had ever developed a set of mathematical equations that could help him figure out the law of warpage, that is, the law that describes how matter forces spacetime to curve. Grossmann, whose specialty was other aspects of